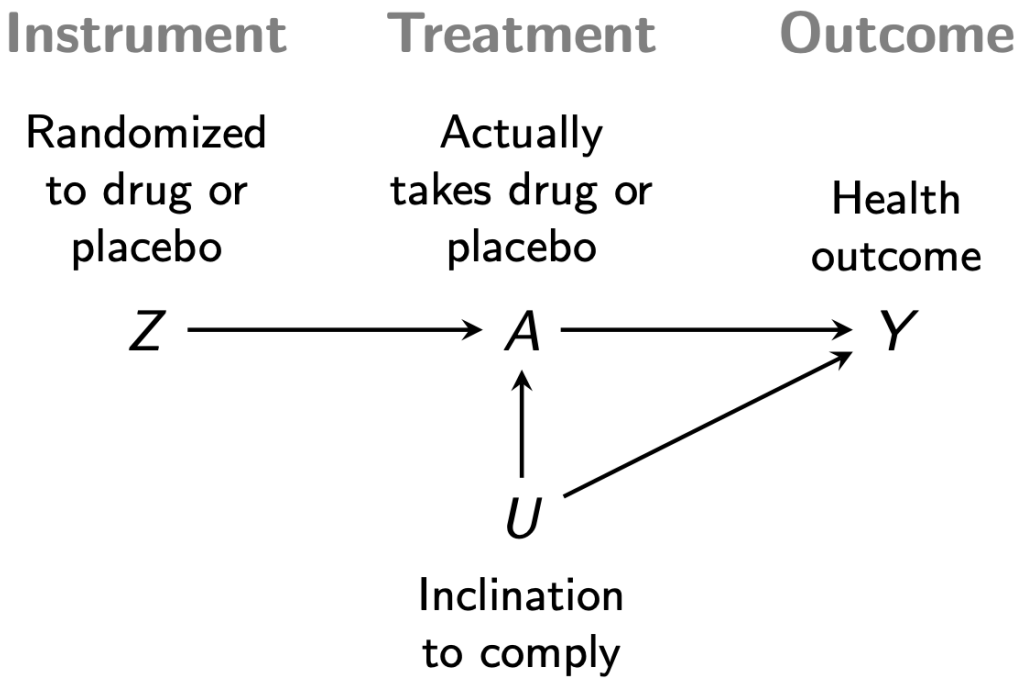
Instrumental Variables

Discussion 9

Instrumental Variables

- Intent to treat effect- the effect of Z on Y
- Average effect among compliers- the effect of Z or A on Y for compliers



Proportion of compliers

$$
\begin{aligned}
&\mathrm{E}(A \mid Z=1) - \mathrm{E}(A \mid Z=0) = \mathrm{E}(A^{Z=1} - A^{Z=0}) \\
&= \sum_s \mathrm{E}(A^{Z=1} - A^{Z=0} \mid S=s) \underbrace{\mathrm{P}(S=s)}_{\substack{\text{Denote} \\ \pi_s}} \\
&= \mathrm{E}(A^{Z=1} - A^{Z=0} \mid S=\text{Complier})\pi_{\text{Complier}} \\
&\quad + \mathrm{E}(A^{Z=1} - A^{Z=0} \mid S=\text{Always-Taker})\pi_{\text{Always-Taker}} \quad (=0) \\
&\quad + \mathrm{E}(A^{Z=1} - A^{Z=0} \mid S=\text{Never-Taker})\pi_{\text{Never-Taker}} \quad (=0) \\
&\quad + \mathrm{E}(A^{Z=1} - A^{Z=0} \mid S=\text{Defier})\pi_{\text{Defier}} \quad (=0) \\
&= \pi_{\text{Complier}}
\end{aligned}
$$

ACE of compliers

Deriving the general case:

$$\mathrm{E}(Y \mid Z=1) - \mathrm{E}(Y \mid Z=0) = \mathrm{E}(Y^{Z=1} - Y^{Z=0})$$

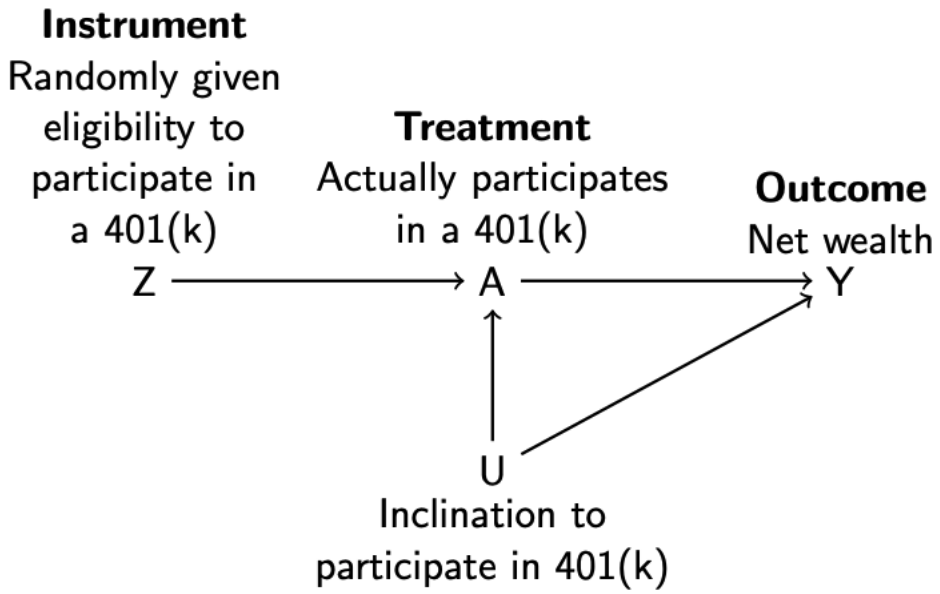
$$= \sum_s \mathrm{E}(Y^{Z=1} - Y^{Z=0} \mid S=s) \underbrace{\mathrm{P}(S=s)}_{\substack{\text{Denote} \\ \pi_s}}$$

$$= \mathrm{E}(Y^{Z=1} - Y^{Z=0} \mid S=\text{Complier})\pi_{\text{Complier}}$$
$$+ \mathrm{E}(Y^{Z=1} - Y^{Z=0} \mid S=\text{Always-Taker})\pi_{\text{Always-Taker}} \quad (=0)$$
$$+ \mathrm{E}(Y^{Z=1} - Y^{Z=0} \mid S=\text{Never-Taker})\pi_{\text{Never-Taker}} \quad (=0)$$
$$+ \mathrm{E}(Y^{Z=1} - Y^{Z=0} \mid S=\text{Defier})\pi_{\text{Defier}} \quad (=0)$$

$$\mathrm{E}(Y^{A=1} - Y^{A=0} \mid S=\text{Complier}) = \frac{\mathrm{E}(Y^{Z=1} - Y^{Z=0})}{\pi_{\text{Complier}}}$$
$$= \frac{\mathrm{E}(Y \mid Z=1) - \mathrm{E}(Y \mid Z=0)}{\mathrm{E}(A \mid Z=1) - \mathrm{E}(A \mid Z=0)}$$

401(k) Example

- Does participating in a 401(k) increase an individual's wealth?
- Participating in a 401(k) is not a random thing!
- However, being eligible for a 401(k) is arguably random.
- 401(k) eligibility affects net wealth *only* through participation.



Two-stage least squares

- When instrument is continuous, the Wald estimator doesn't work
- Two-stage least square steps:
  - Use Z_i to predict $\hat{A}_i$
  - Exclusion: the variation in $\hat{A}_i$ is only because of Z_i and not U_i
  - Regress Y_i onto $\hat{A}_i$
  - Linear coefficient is the estimate of $\hat{A} \rightarrow Y$

- Same as Wald estimate if instrument and treatment are binary

Let’s do it ourselves!

- There is a short coding exercise on the website…